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Architecture Deep-Dive

> Technical walkthrough of every component in the `data_analysis_lg` project — how data flows, how agents are built, how tools are wired, and how the system stays within LLM token limits.

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1. Conceptual Layers

The project is organized into four conceptual layers, each with a clear responsibility:

```
Parse error on line 2:
...ph TB    subgraph "Layer 4 – Orchestrat...
----------------------^
Expecting 'SEMI', 'NEWLINE', 'SPACE', 'EOF', 'GRAPH', 'DIR',
'TAGEND', 'TAGSTART', 'UP', 'DOWN', 'subgraph', 'end', 'SQE',
'PE', '-)', 'DIAMOND_STOP', 'MINUS', '--', 'ARROW_POINT',
'ARROW_CIRCLE', 'ARROW_CROSS', 'ARROW_OPEN',
'DOTTED_ARROW_POINT', 'DOTTED_ARROW_CIRCLE',
'DOTTED_ARROW_CROSS', 'DOTTED_ARROW_OPEN', '==',
'THICK_ARROW_POINT', 'THICK_ARROW_CIRCLE', 'THICK_ARROW_CROSS',
'THICK_ARROW_OPEN', 'PIPE', 'STYLE', 'LINKSTYLE', 'CLASSDEF',
'CLASS', 'CLICK', 'DEFAULT', 'NUM', 'PCT', 'COMMA', 'ALPHA',
```

```
'COLON', 'BRKT', 'DOT', 'PUNCTUATION', 'UNICODE_TEXT', 'PLUS',
'EQUALS', 'MULT', got 'STR'
```

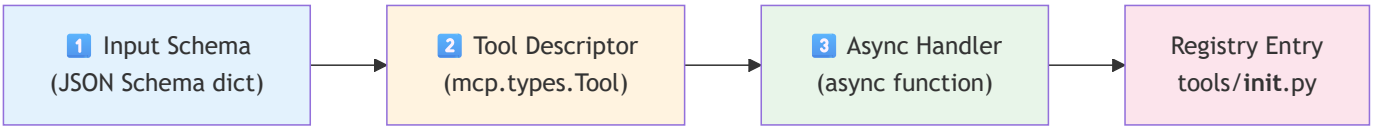
Layer	Files	Responsibility
Layer 1 — Tool Execution	`server.py`, `tools/*`	Host MCP tools, execute data operations
Layer 2 — Protocol Bridge	`client.py`	Convert MCP tools ↔ LangChain StructuredTools
Layer 3 — Agent Intelligence	`graph.py`	Define agents, prompts, routing, state machine
Layer 4 — Orchestration	`main.py`	Wire everything together, run analysis

---

2. The MCP Tool Server

2.1 Tool Pattern (Three-Part Convention)

Every tool in this project follows the exact same pattern inherited from the `lg_mcp_01` reference project. This consistency makes adding new tools trivial.



Here's the pattern applied to `load_csv` as an example:

Part 1 — Input Schema (defines what the LLM sees as parameters):

```
load_csv_input_schema = {
    "type": "object",
    "properties": {
        "file_path": {
            "type": "string",
            "description": "Path to the CSV file, relative to csv/ folder or
absolute.",
        },
        "nrows": {
            "type": "integer",
            "description": "Optional: max rows to load. Omit to load all rows.",
```

```
        },
    },
    "required": ["file_path"],  # nrows is optional
}
```

Part 2 — Tool Descriptor (MCP metadata for discovery):

```
load_csv_tool = types.Tool(
    name="load_csv",
    description="Load a CSV file into the server's in-memory dataframe store...",
    inputSchema=load_csv_input_schema,
)
```

Part 3 — Async Handler (does the actual work):

```
async def load_csv(arguments: dict) -> list[types.TextContent]:
    file_path = arguments.get("file_path", "").strip()
    nrows = arguments.get("nrows", 0)
    # ... load data, return JSON result
    return [types.TextContent(type="text", text=json.dumps(result, default=str))]
```

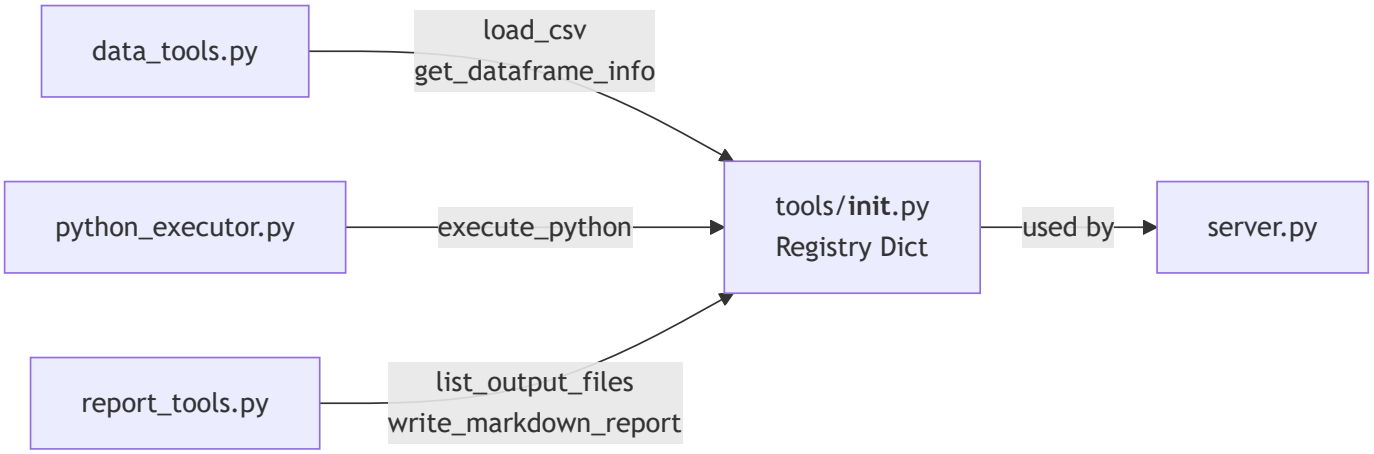
> **Adding a new tool** requires only: (1) create the three parts in a file under `tools/`, (2) add one entry in `tools/__init__.py`. No changes to `server.py` or `client.py`.

2.2 Tool Registry

The `tools/__init__.py` file acts as a central catalog. The server iterates this dict to respond to `list_tools` and `call_tool` requests:

```
# tools/__init__.py
tools = {
    load_csv_tool.name:              {"tool": load_csv_tool,
"handler": load_csv},
    get_dataframe_info_tool.name:    {"tool": get_dataframe_info_tool,
"handler": get_dataframe_info},
    execute_python_tool.name:        {"tool": execute_python_tool,
"handler": execute_python},
    list_output_files_tool.name:     {"tool": list_output_files_tool,
"handler": list_output_files},
    write_markdown_report_tool.name: {"tool": write_markdown_report_tool,
```

```
    "handler": write_markdown_report},
    }
```



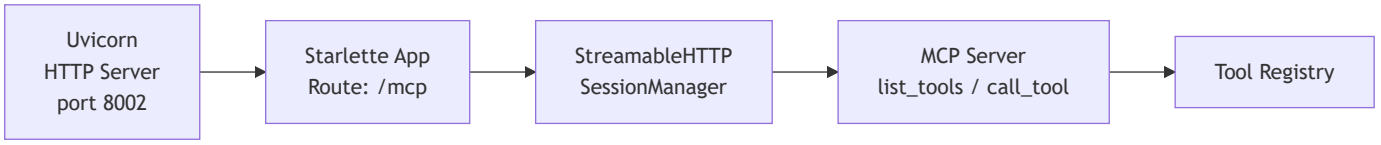
2.3 Server Wiring

`server.py` sets up the MCP server with two handler functions that delegate everything to the registry:

```
@server.list_tools()
async def handle_list_tools() -> list[types.Tool]:
    return [entry["tool"] for entry in tools.values()]

@server.call_tool()
async def handle_call_tool(name: str, arguments: dict) -> list[types.TextContent]:
    if name not in tools:
        raise ValueError(f"Unknown tool: {name}")
    handler = tools[name]["handler"]
    return await handler(arguments)
```

The transport stack is:

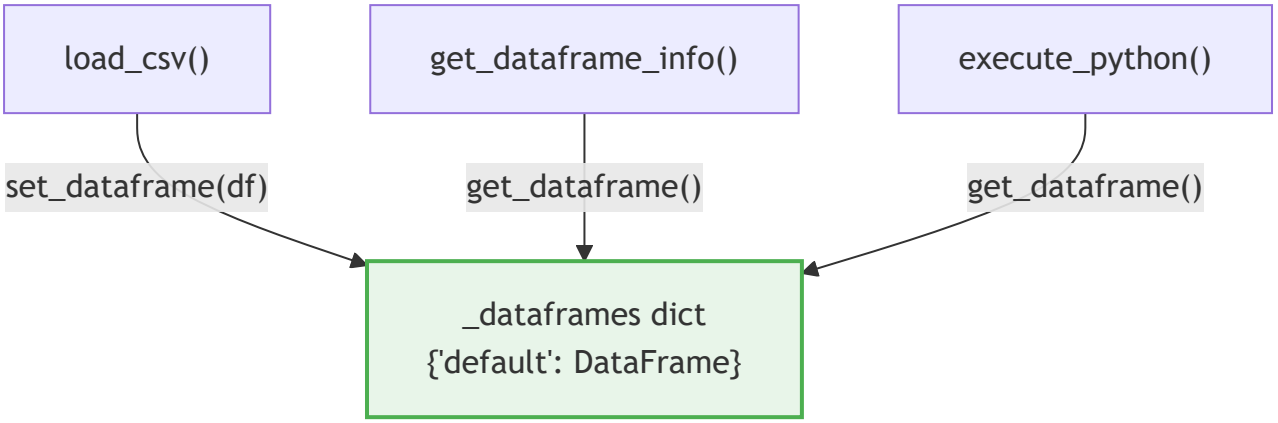
```
Tool Handlers → MCP Server → StreamableHTTPSessionManager → Starlette App → Uvicorn
(port 8002)
```



2.4 In-Memory DataFrame Store

The DataFrame is stored in a module-level dictionary in `data_tools.py`. This is a simple but effective approach for a single-server demo:

```
# Shared across all tools in this server process
_dataframes: dict[str, pd.DataFrame] = {}

def get_dataframe(name: str = "default") -> pd.DataFrame | None:
    return _dataframes.get(name)

def set_dataframe(df: pd.DataFrame, name: str = "default") -> None:
    _dataframes[name] = df
```



Why a dict with a `name` key? This allows future multi-dataset support (e.g., loading two CSVs and joining them). Currently only `"default"` is used.

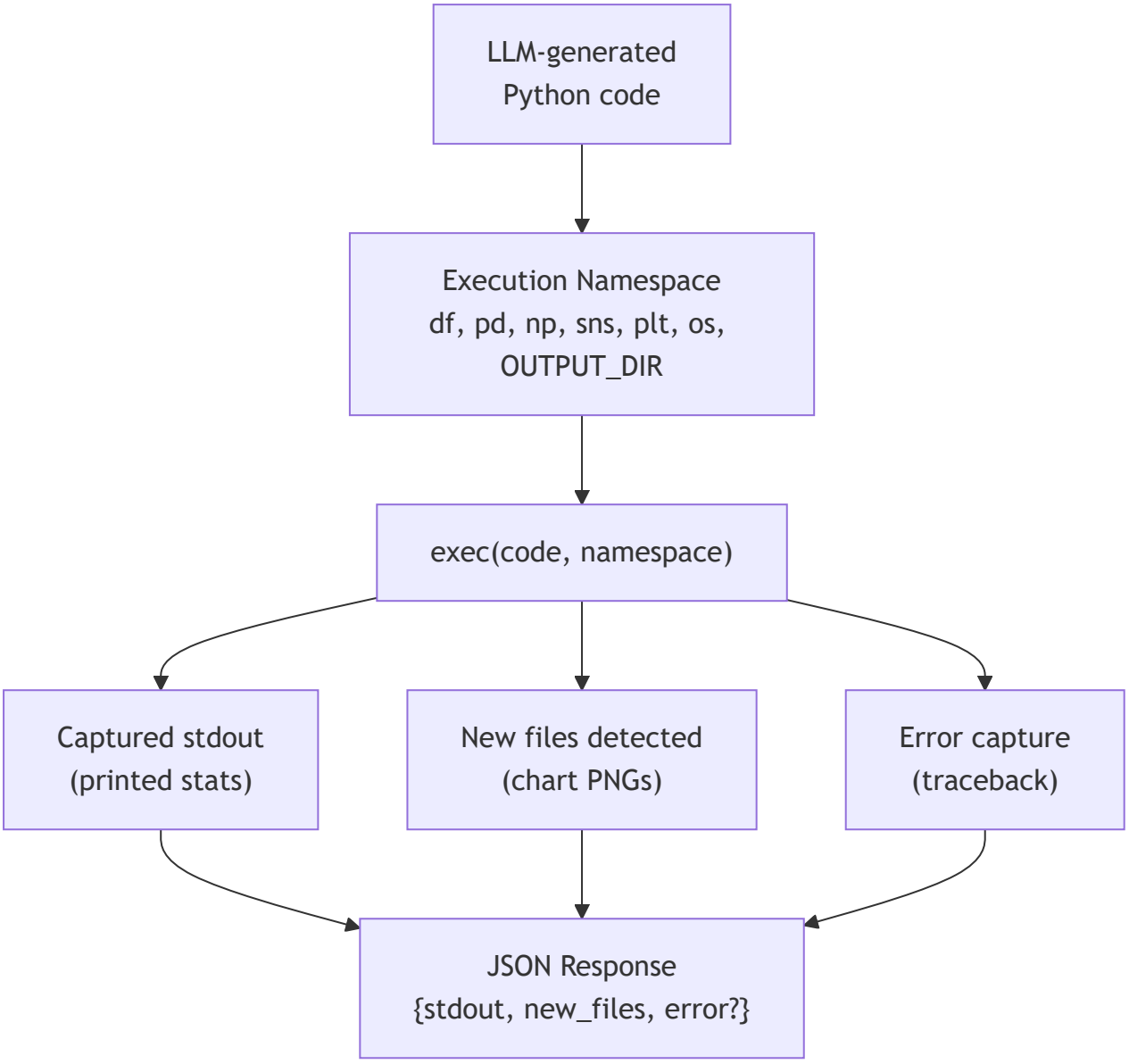
2.5 Python Executor: Sandboxed Code Execution

The `execute_python` tool is the most complex tool. It runs arbitrary Python code from the LLM in a controlled namespace:

```
# Build execution namespace
namespace = {
    "df": df.copy(),          # Copy to prevent mutation of the shared store
    "pd": pd, "np": np,
    "sns": sns, "plt": plt,
    "os": os,
    "OUTPUT_DIR": OUTPUT_DIR,
    "__builtins__": __builtins__,
```

```
    }

    # Capture stdout
    old_stdout = sys.stdout
    sys.stdout = captured_stdout = io.StringIO()

    try:
        exec(code, namespace)
    except Exception:
        error_msg = traceback.format_exc()
    finally:
        sys.stdout = old_stdout
        plt.close("all")       # Free memory from any unclosed figures
```

Key features:



Feature	Implementation	Why
DataFrame copy	`df.copy()`	Prevents LLM code from corrupting the shared store

Feature	Implementation	Why
stdout capture	`io.StringIO` redirect	Agents see `print()` output from executed code
File detection	`set(os.listdir(OUTPUT_DIR))` diff	Tracks which charts were saved
Figure cleanup	`plt.close("all")` in `finally`	Prevents memory leaks from matplotlib
Error handling	`traceback.format_exc()`	Agent sees the error and can retry
Output truncation	`stdout[:5000]`, `error[:3000]`	Prevents huge outputs from blowing up context

> **Security note:** This is a demo. In production, use a proper sandbox (e.g., Docker container, gVisor, or a dedicated code execution service).

---

3. The MCP-to-LangChain Bridge

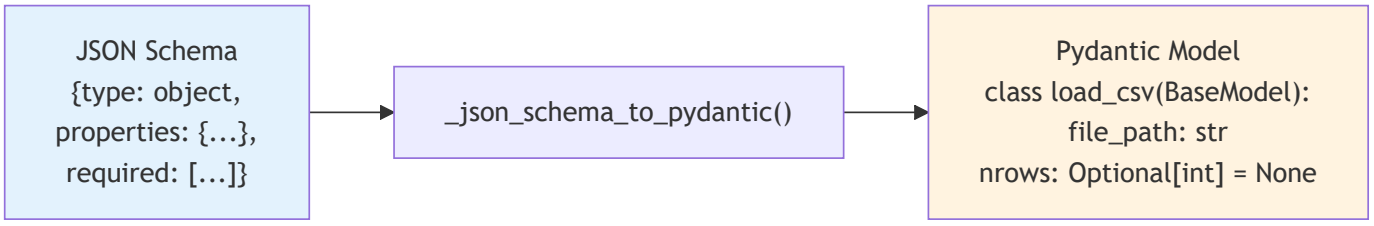
`client.py` solves a critical integration problem: **MCP tools use JSON Schema**, but **LangGraph's ToolNode needs Pydantic models**. The bridge converts between the two worlds.

3.1 JSON Schema → Pydantic Conversion

```
def _json_schema_to_pydantic(schema: dict, model_name: str) -> Type[BaseModel]:
    properties = schema.get("properties", {})
    required = set(schema.get("required", []))

    field_definitions: dict[str, Any] = {}
    for prop_name, prop_meta in properties.items():
        json_type = prop_meta.get("type", "string")
        python_type = int if json_type == "integer" else str

        if prop_name in required:
            field_definitions[prop_name] = (python_type, ...)       # Required
        else:
            field_definitions[prop_name] = (Optional[python_type], None)  #
Optional
```

```
    return create_model(model_name, **field_definitions)
```

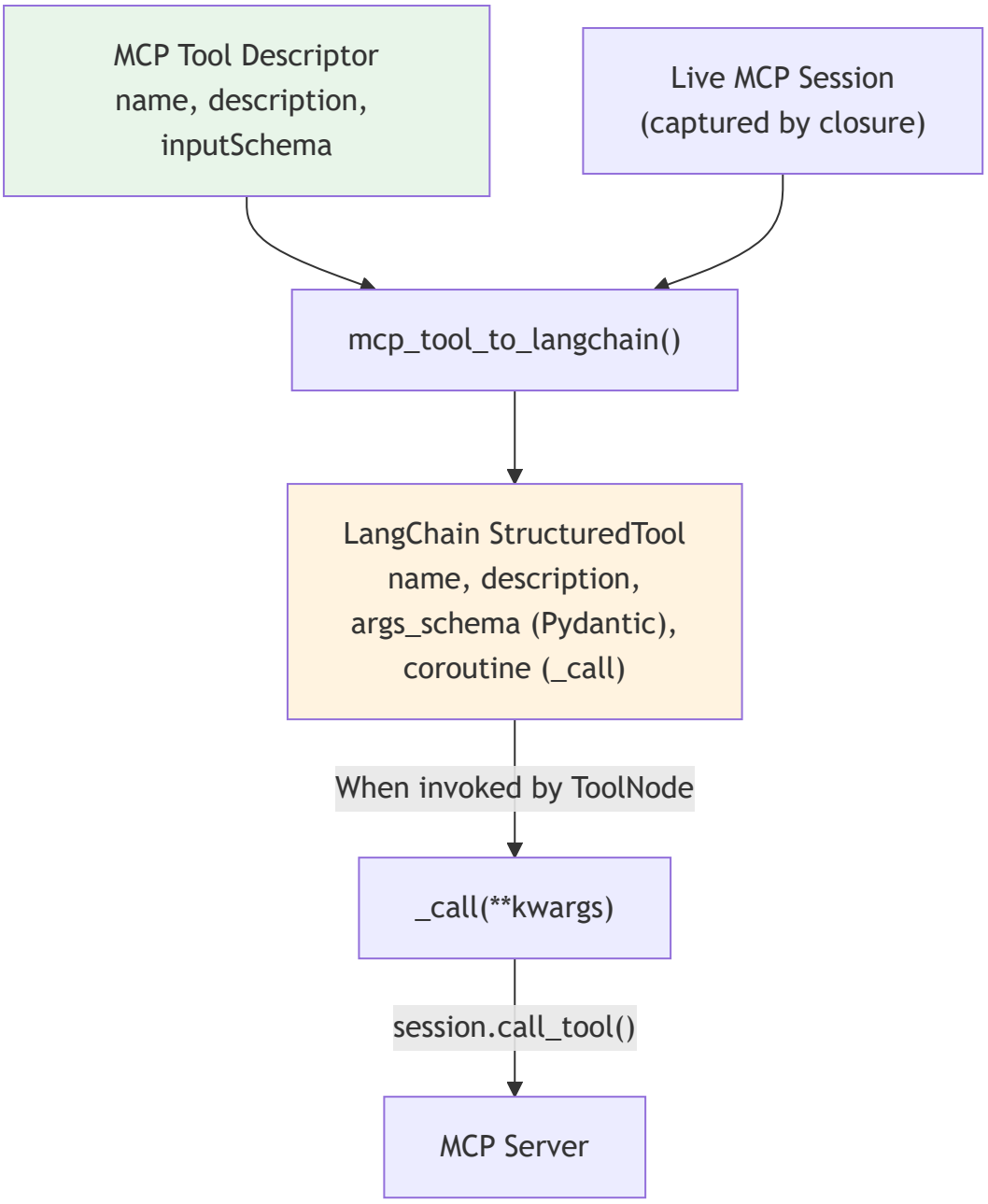
This dynamically builds Pydantic models at runtime:



3.2 Tool Wrapping with Closure

Each MCP tool is wrapped as a LangChain `StructuredTool` using a closure that captures the live MCP session:

```
def mcp_tool_to_langchain(tool: mcp_types.Tool, session: ClientSession) ->
BaseTool:
    args_schema = _json_schema_to_pydantic(tool.inputSchema, model_name=tool.name)

    async def _call(**kwargs: Any) -> str:
        clean_kwargs = {k: v for k, v in kwargs.items() if v is not None}
        result = await session.call_tool(name=tool.name, arguments=clean_kwargs)
        if result.isError:
            return f"Error: {result.content[0].text}"
        return result.content[0].text

    return StructuredTool.from_function(
        coroutine=_call,
        name=tool.name,
        description=tool.description,
        args_schema=args_schema,
    )
```



3.3 Handling Optional / Nullable Parameters

A subtle but important fix: LLMs often send `null` for optional parameters (e.g., `{"nrows": null}`). The Groq API validates strictly against the JSON schema and rejects `null` for `integer` types. Two fixes work together:

1. **Pydantic side:** Optional fields use `Optional[int]` so the schema allows `null`
2. **Call side:** `None` values are filtered out before sending to the MCP server

```
# In _call closure:
clean_kwargs = {k: v for k, v in kwargs.items() if v is not None}
```

This prevents the error: `tool call validation failed: parameters for tool load_csv did not match schema: /nrows: expected integer, but got null`.

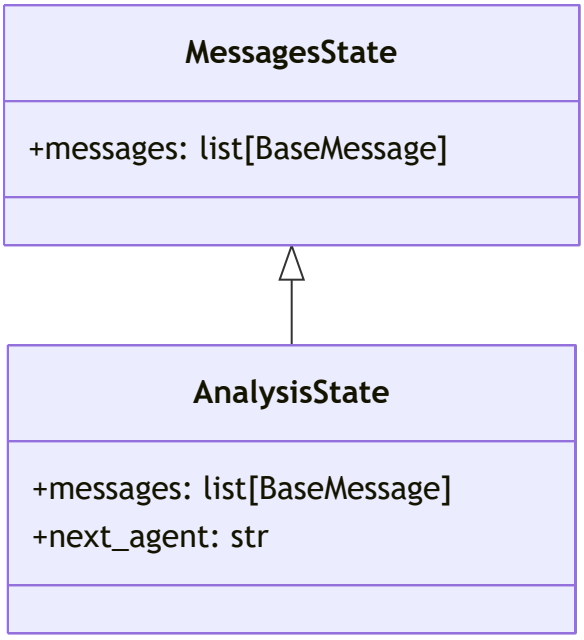
---

4. The Multi-Agent Graph

4.1 Custom State: AnalysisState

LangGraph's built-in `MessagesState` provides a shared message list. We extend it with a `next_agent` field so the Manager can signal routing decisions:

```
class AnalysisState(MessagesState):
    next_agent: str
```



The `messages` list is **append-only** — every agent adds its output, and tool results are automatically inserted by `ToolNode`. This gives each agent full context of what has happened so far (subject to compression, see 4.6).

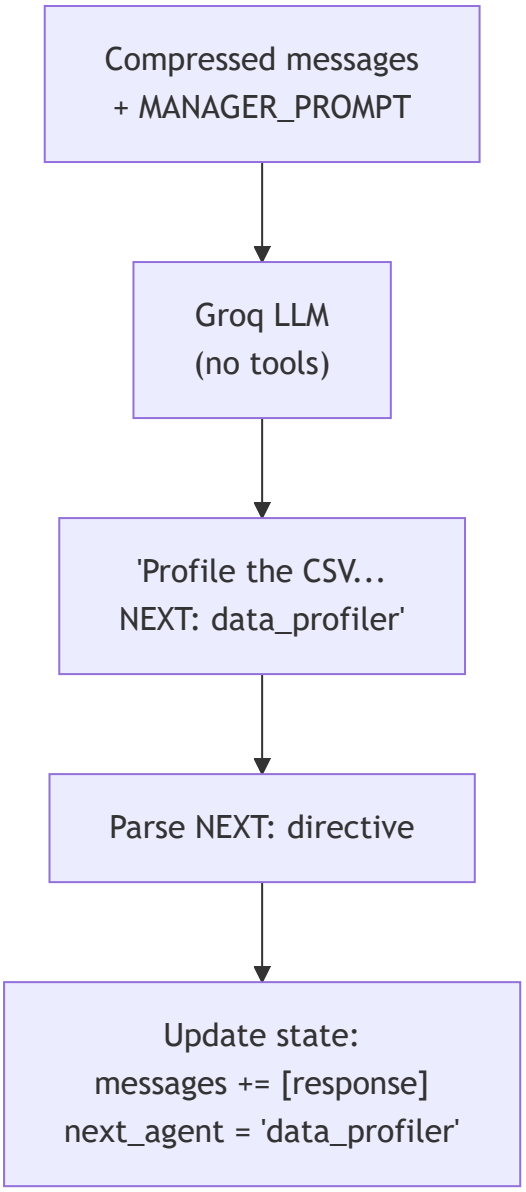
4.2 The Manager Node

The Manager is the only agent that **does not use tools**. It reads the conversation and emits a routing directive:

```python
def manager_node(state: AnalysisState) -> dict:
    messages = [SystemMessage(content=MANAGER_PROMPT)] +
compress_messages(state["messages"])
    response = llm.invoke(messages)  # No tools bound

    # Parse the NEXT: directive from the response
    next_agent = "FINISH"
    for line in content.strip().split("\n"):
        if line.strip().startswith("NEXT:"):
            next_agent = line.split("NEXT:")[1].strip()
            break

    return {
        "messages": [AIMessage(content=content, name="manager")],
        "next_agent": next_agent,
    }
```

The Manager's system prompt enforces a strict output format:

```
Respond with a 1-2 sentence instruction, then on its own line:
NEXT: <agent_name>
```



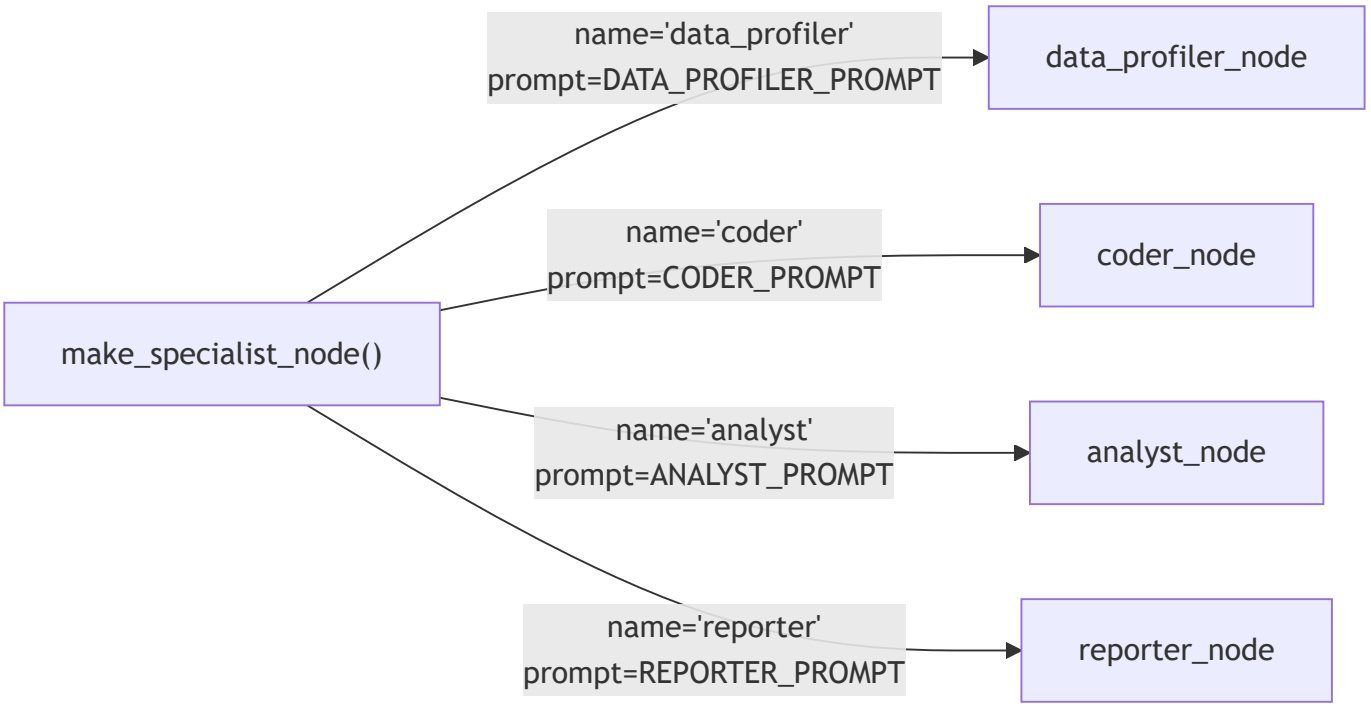
4.3 Specialist Agent Factory

All four specialist agents are created by the same factory function, differentiated only by their system prompt:

```
def make_specialist_node(name: str, system_prompt: str):
    def specialist_node(state: AnalysisState) -> dict:
        iteration = sum(1 for m in state["messages"] if getattr(m, "name", None) ==
name) + 1
        messages = [SystemMessage(content=system_prompt)] +
compress_messages(state["messages"])
        response = llm_with_tools.invoke(messages)   # Tools ARE bound here
        response.name = name
        return {"messages": [response]}
    return specialist_node
```

Key differences from the Manager:

- Uses `llm_with_tools` (tools are bound to the LLM)
- Does **not** set `next_agent` — routing back to Manager is handled by conditional edges
- Tags each response with `response.name = name` for tracking



4.4 Routing Logic (Three Router Functions)

The graph uses three routing functions for conditional edges:

Router 1: `manager_router` — Where does the Manager send work?

```
def manager_router(state: AnalysisState) -> str:
    next_agent = state.get("next_agent", "FINISH")
    if next_agent in ("data_profiler", "coder", "analyst", "reporter"):
        return next_agent
    return "end"
```

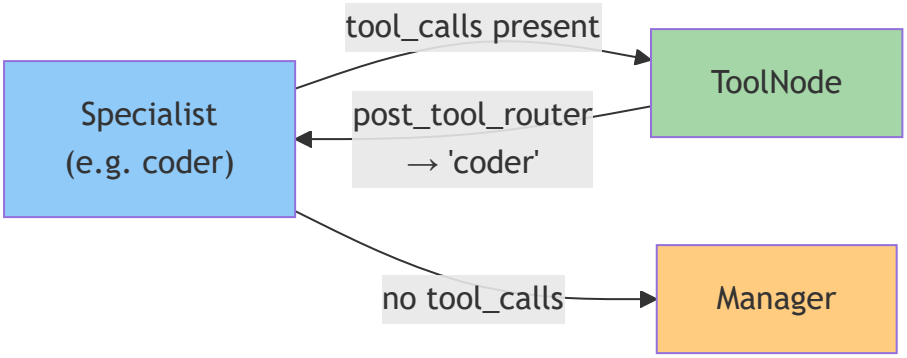
Router 2: `specialist_router` — Does the specialist need tools?

```
def specialist_router(state: AnalysisState) -> str:
    last_message = state["messages"][-1]
    if hasattr(last_message, "tool_calls") and last_message.tool_calls:
        return "tools"
    return "manager"      # Done — hand back to manager
```

Router 3: `post_tool_router` — After tools run, who called them?

```
def post_tool_router(state: AnalysisState) -> str:
    for msg in reversed(state["messages"]):
        if hasattr(msg, "tool_calls") and msg.tool_calls and getattr(msg, "name",
None):
            return msg.name   # Return to the specialist that made the call
    return "manager"
```

Together these create the **specialist ↔ tools loop**:



A specialist can call tools **multiple times** in a row. For example, the Data Profiler typically:

1. Calls `load_csv` → gets routed back → sees result
2. Calls `get_dataframe_info` → gets routed back → sees result
3. Writes a summary (no tool_calls) → routes to Manager

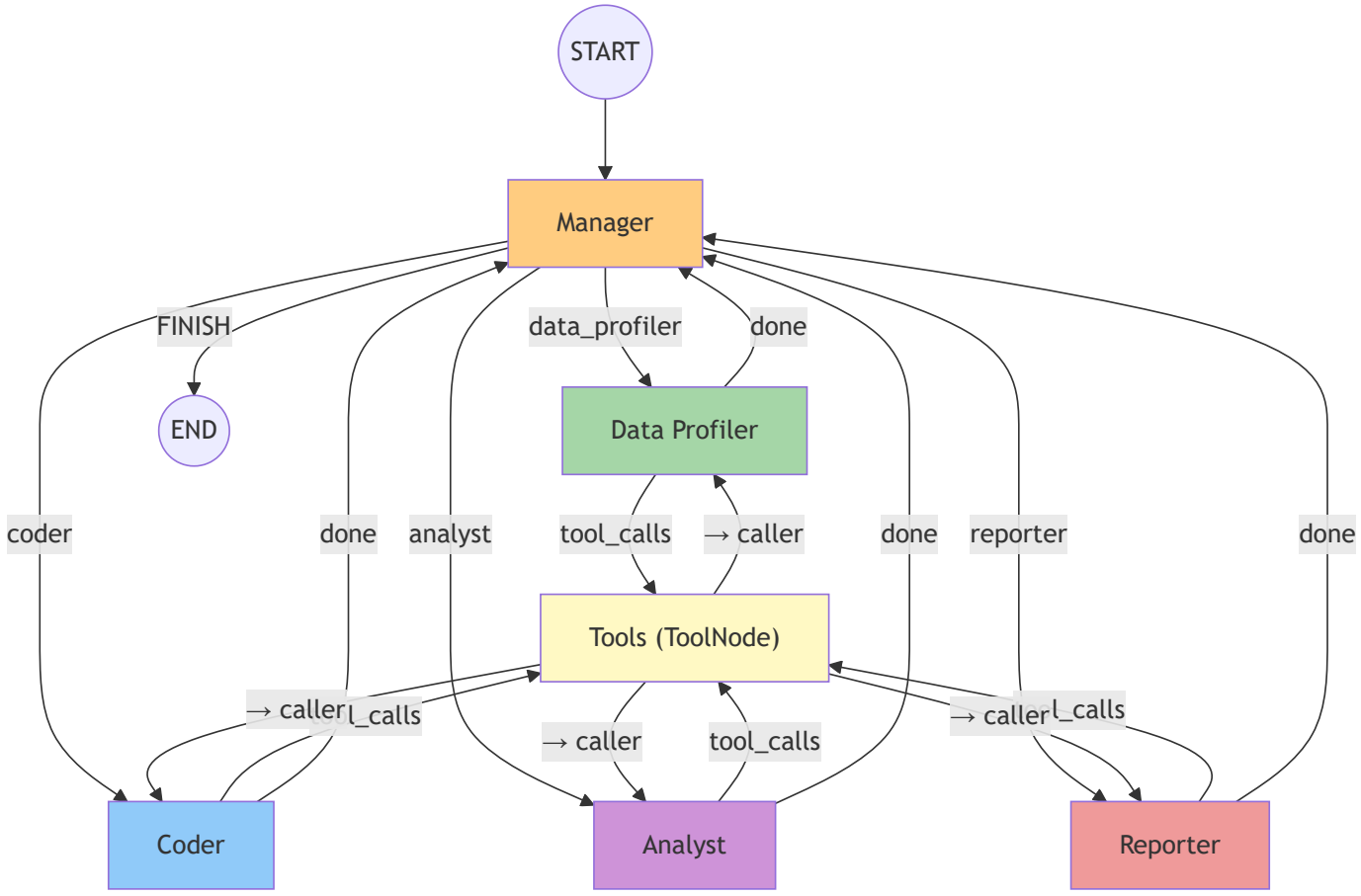
4.5 Graph Assembly

The full graph is assembled with `StateGraph`:

```
graph_builder = StateGraph(AnalysisState)

# Register all nodes
graph_builder.add_node("manager", manager_node)
graph_builder.add_node("data_profiler", data_profiler_node)
graph_builder.add_node("coder", coder_node)
graph_builder.add_node("analyst", analyst_node)
graph_builder.add_node("reporter", reporter_node)
graph_builder.add_node("tools", tool_node)

# Entry point
graph_builder.set_entry_point("manager")
```

```python
# Manager → specialist or END
graph_builder.add_conditional_edges("manager", manager_router, {
    "data_profiler": "data_profiler",
    "coder": "coder",
    "analyst": "analyst",
    "reporter": "reporter",
    "end": END,
})

# Each specialist → tools or manager
for specialist in ["data_profiler", "coder", "analyst", "reporter"]:
    graph_builder.add_conditional_edges(specialist, specialist_router, {
        "tools": "tools",
        "manager": "manager",
    })

# Tools → back to calling specialist
graph_builder.add_conditional_edges("tools", post_tool_router, {
    "data_profiler": "data_profiler",
    "coder": "coder",
    "analyst": "analyst",
    "reporter": "reporter",
    "manager": "manager",
})
```

Full compiled graph visualization:

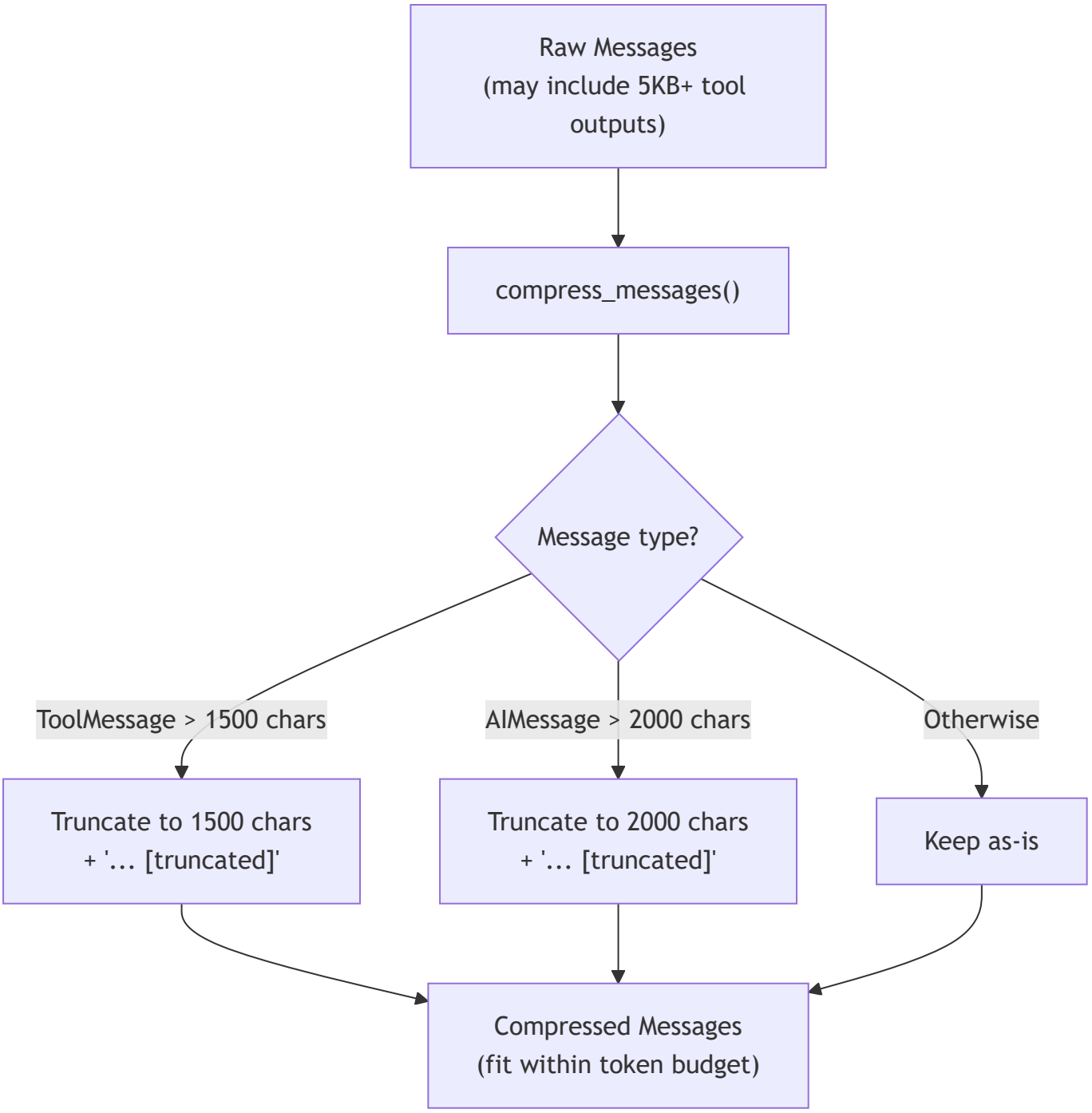


4.6 Message Compression (Token Management)

The Groq free tier has an **8,000 tokens-per-minute (TPM)** limit. Without compression, the conversation quickly exceeds this as tool results accumulate (e.g., DataFrame info with 20+ columns, code execution stdout).

The `compress_messages()` function truncates large messages before sending them to the LLM:

```
MAX_TOOL_OUTPUT_CHARS = 1500

def compress_messages(messages: list[BaseMessage]) -> list[BaseMessage]:
    compressed = []
    for msg in messages:
        if isinstance(msg, ToolMessage):
            content = msg.content if isinstance(msg.content, str) else
str(msg.content)
            if len(content) > MAX_TOOL_OUTPUT_CHARS:
                truncated = content[:MAX_TOOL_OUTPUT_CHARS] + "\n... [truncated]"
                compressed.append(ToolMessage(
                    content=truncated,
                    tool_call_id=msg.tool_call_id,
                    name=msg.name,
                ))
            else:
                compressed.append(msg)
        elif isinstance(msg, AIMessage) and msg.content and len(msg.content) >
2000:
            truncated_msg = AIMessage(
                content=msg.content[:2000] + "\n... [truncated]",
                name=getattr(msg, "name", None),
                tool_calls=msg.tool_calls if msg.tool_calls else [],
            )
            compressed.append(truncated_msg)
        else:
            compressed.append(msg)
    return compressed
```



Where is compression applied?

- ✅ Before every Manager LLM call
- ✅ Before every Specialist LLM call
- ❌ NOT on the actual state (original messages are preserved for downstream agents)

This means the **state retains full fidelity**, but each LLM call only sees a compressed view.

---

5. The Entry Point (main.py)

`main.py` wires everything together in a clean async pipeline:

```
async def main() -> None:
    api_key = os.getenv("GROQ_API_KEY")

    async with streamable_http_client(SERVER_URL) as (read, write, _):
        async with ClientSession(read, write) as session:
            await session.initialize()

            # Fetch tools once, build graph once
            langchain_tools = await get_mcp_tools(session)
            graph = build_graph(langchain_tools, api_key)

            # Run each demo analysis
            for label, user_input in DEMO_REQUESTS:
                await run_analysis(graph, label, user_input)
```

The `DEMO_REQUESTS` list defines analysis scenarios to run:

```
DEMO_REQUESTS = [
    (
        "Correlation Analysis",
        "Calculate the correlation of the 'Ore' and 'DensityHC' columns...",
    ),
]
```

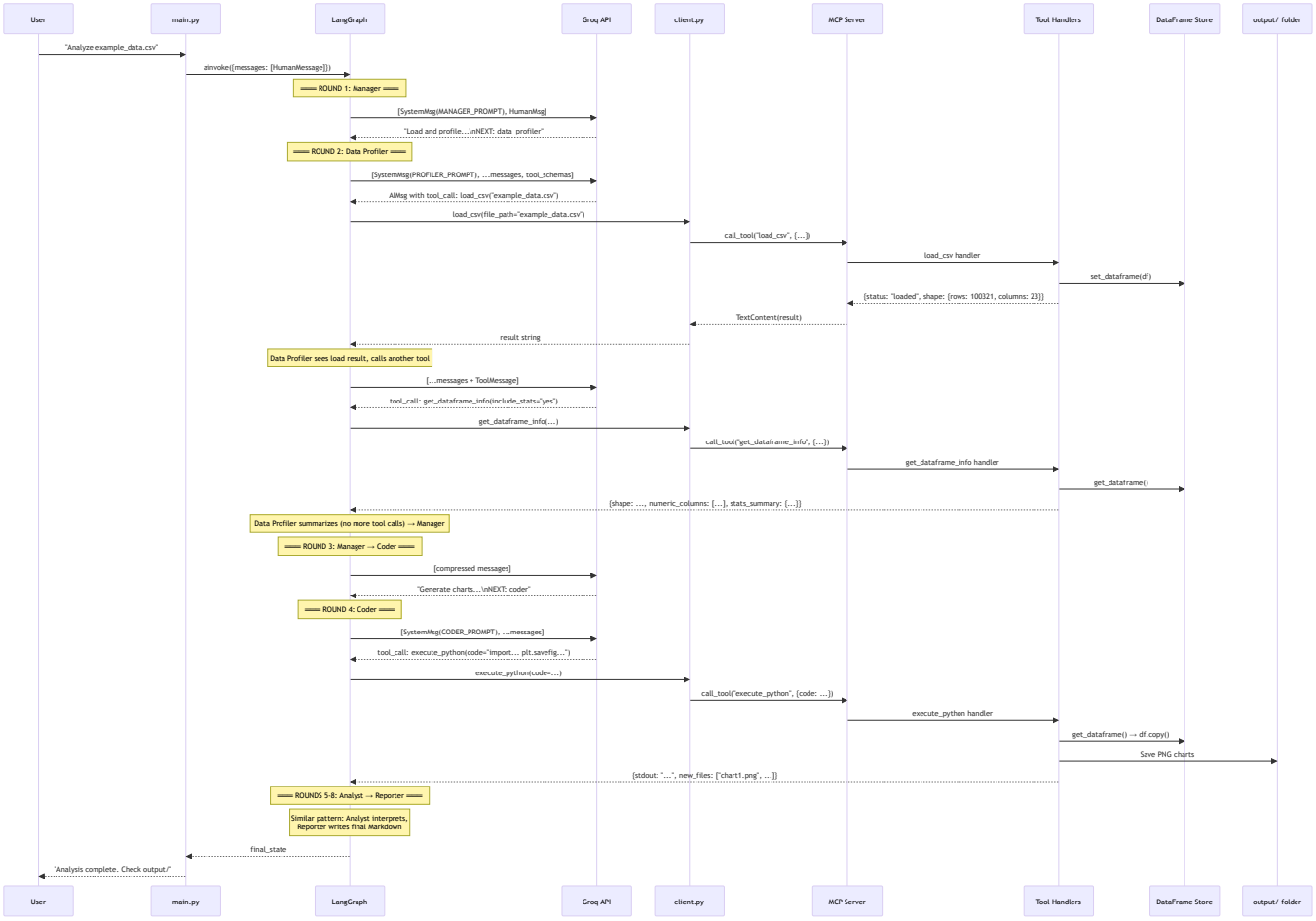
Each request is invoked with:

```
final_state = await graph.ainvoke(
    {"messages": [HumanMessage(content=user_input)]},
    config={
        "configurable": {"thread_id": label},
        "recursion_limit": 50,
    },
)
```

---

6. Data Flow: End-to-End Trace

Here's a complete trace of a typical analysis request flowing through the system:



7. Design Decisions & Trade-offs

Decision	Rationale	Alternative Considered
Two-process architecture	Follows MCP spec; tools could run on a different machine	Single-process with direct function calls
Manager as router (not LLM tool)	Simpler parsing; avoids tool-call overhead for routing	LangGraph's built-in `Command` for routing
Factory pattern for specialists	DRY — same logic, different prompts	Separate functions per agent
Message compression	Required for Groq free tier (8K TPM limit)	Summarization via separate LLM call (more expensive)
`exec()` for code execution	Fast, direct access to DataFrame	Docker container (safer but slower)

Decision	Rationale	Alternative Considered
In-memory DataFrame dict	Simple; works for single-user demo	Database or file-based store for multi-user
Compact tool outputs	Keep token count low per interaction	Full outputs with aggressive summarization
`NEXT:` **directive parsing**	Simple string parsing; works reliably	Structured output / function call for routing
Single `ToolNode` **shared by all agents**	All agents share the same tool set	Per-agent tool subsets (more restrictive)
`df.copy()` **in executor**	Prevents LLM code from corrupting shared state	Copy-on-write (more complex)